Instrumentation, Data analysis and Machine Learning

Master in Astrophysics, Particle Physics and Cosmology

P. Bordas, X. Luri, C. Marin

ICCUB
Institute of Cosmos Sciences
EXCELENCIA MARÍA DE MAEZTU
UNIVERSITAT DE BARCELONA

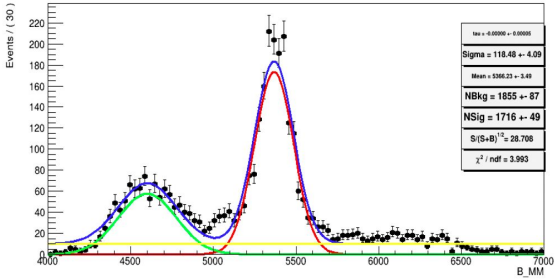
Content

Instrumentation

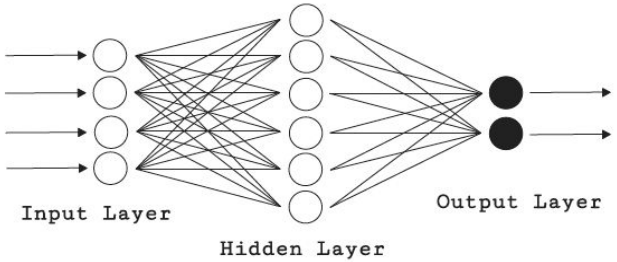
Data analysis

Machine learning

Joan Oro telescope (CAM)



Credit: A. Santander



Credit: V. Costa

General information

- Optional subject (2nd semester), joint astro & particle & cosmology
  - 6 credits = 150h
  - Lectures: 12h (first weeks)
  - Practical sessions: 20h
  - Workshops and field trips: 20h
  - Supervised project: 50h
  - Independent learning: 48h

- Professors
  - Astrophysics: P. Bordas and X. Luri
  - Particle physics: C. Marin

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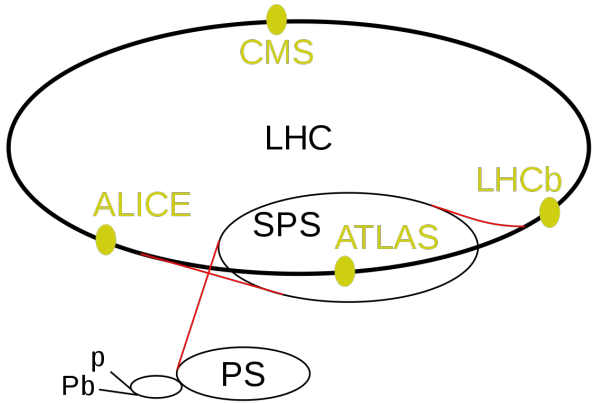
Dpt. Física Quàntica i Astrofísica

Lectures

First weeks (12h):

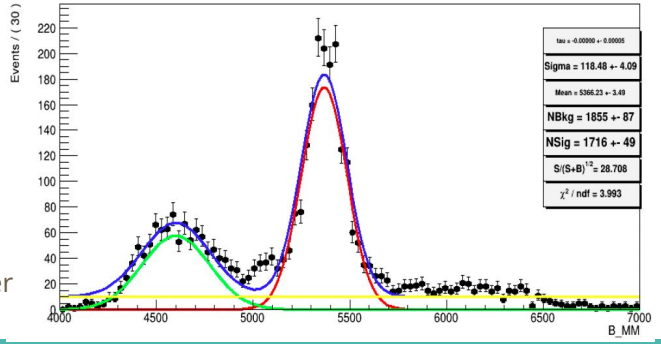
1. Requirements of particle physics experiments
2. Requirements of astrophysical observations
3. Particle accelerators
4. Detection techniques
5. Design of high energy physics experiments
6. Data acquisition and processing
7. Astrophysical instrumentation
8. Astrophysical observation techniques



Practical sessions

During all the course (20h):

- Machine learning and fitting techniques using LHCb data
- Cloud computing using Gaia data
- X-rays using Chandra and XMM-Newton data
- High-Energy gamma rays using Fermi-LAT data
- Very high-energy gamma rays using CTA data



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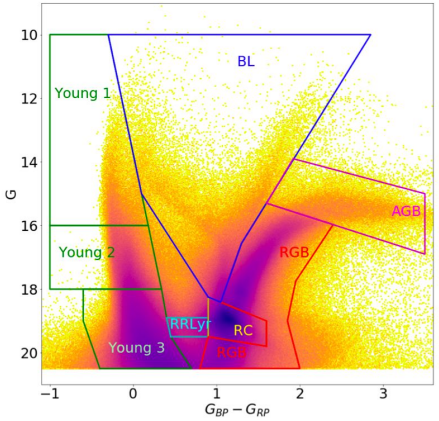
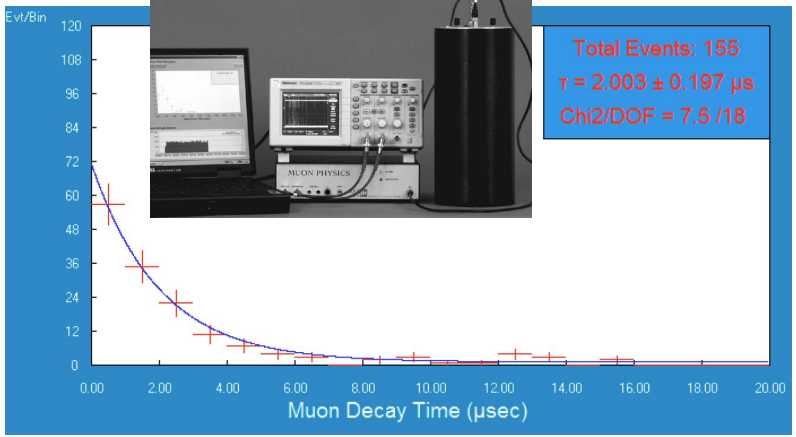


Workshops and field trips (1)

During all the course (20h):

ALBA synchrotron beam emittance

Muon lifetime



Credit: J. R. Stocchetti

Workshops and field trips (2)

OPTIONAL

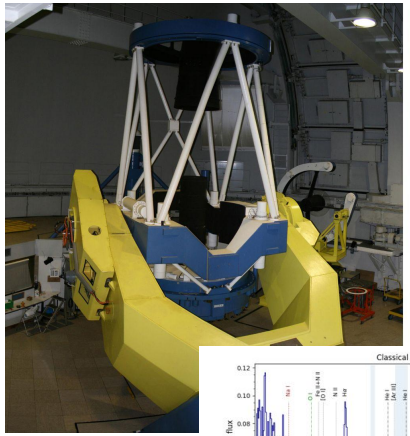
During all the course (20h):

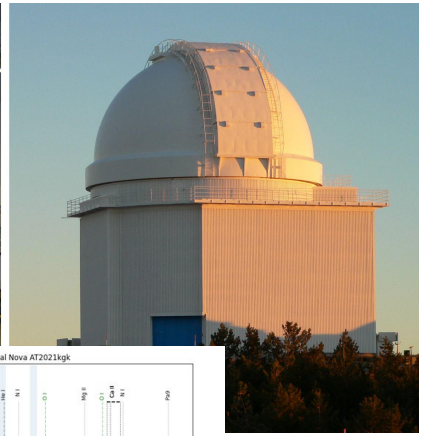
Observation at

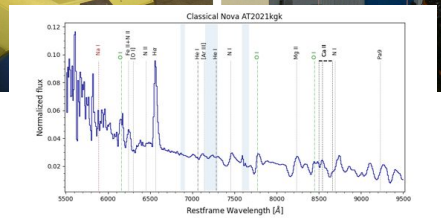
Observatori Astronòmic del Montsec

Proposal and observation at

Calar Alto Astronomical Observatory





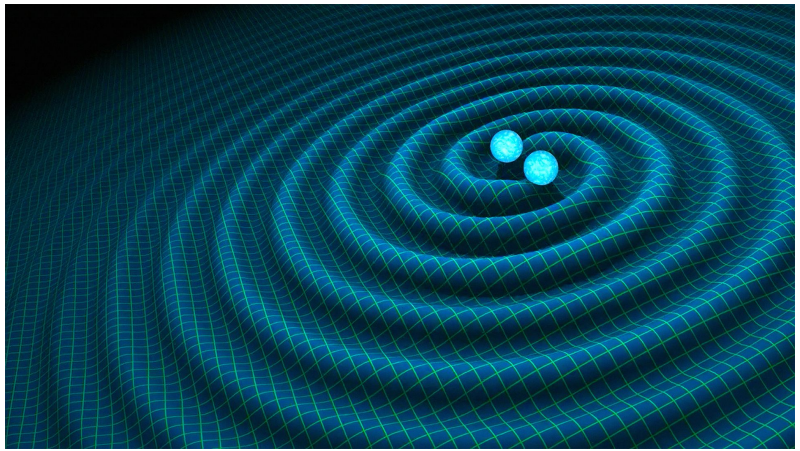


Extinction-corrected spectrum of the galactic nova AT2021kgk obtained with instrument CAFOS attached to the 2.2 m Zeiss reflector of Calar Alto, in observing time of the Calar Alto Academy program assigned to practical work for the University of Barcelona. Credits: Nadia Blagorodovna.

New this year: Gravitational waves (optional)

- Topic already covered in astrophysical instrumentation lecture
- Dedicated Scientific Seminar (1h) this course
- Topics will be proposed for final supervised project using LIGO-Virgo data

by T. Andrade and O. Bulashenko



Supervised project (50h)

To be submitted at the end of the course (start early!)

Related to a topic of the subject or to your TFM project (extension)

Topic examples:

- Related to site testing, instrument characterisation or telescope setting up
- Calibration or time alignment of detectors
- LHCb, Fermi satellite, MAGIC telescope, Gaia or LIGO-Virgo data analysis
- Radio interferometric observations

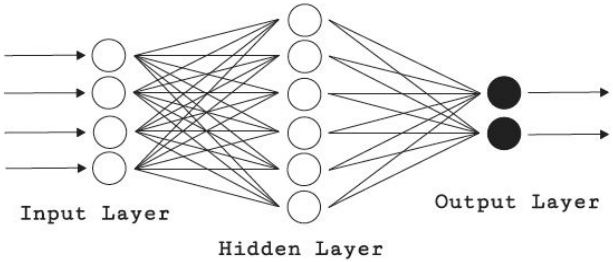
Examples from last course:

- *Processing and stacking astrophotography images using python and DSS*
- *Boosted Decision Tree implementation in Lepton Universality studies in* $B_s \rightarrow D_s^{**} l \nu$

Summary

Unique opportunity to access instrumentation and data from ongoing Particle Physics and Astrophysics experiments!

Joan Oro telescope (CAM)



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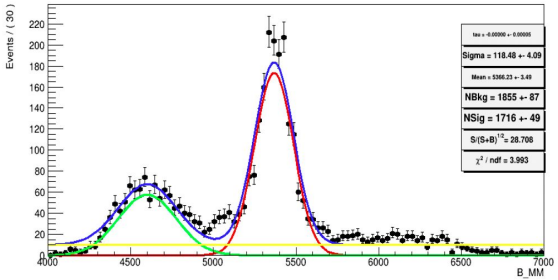

Credit: A. Santander

Thanks for the attention